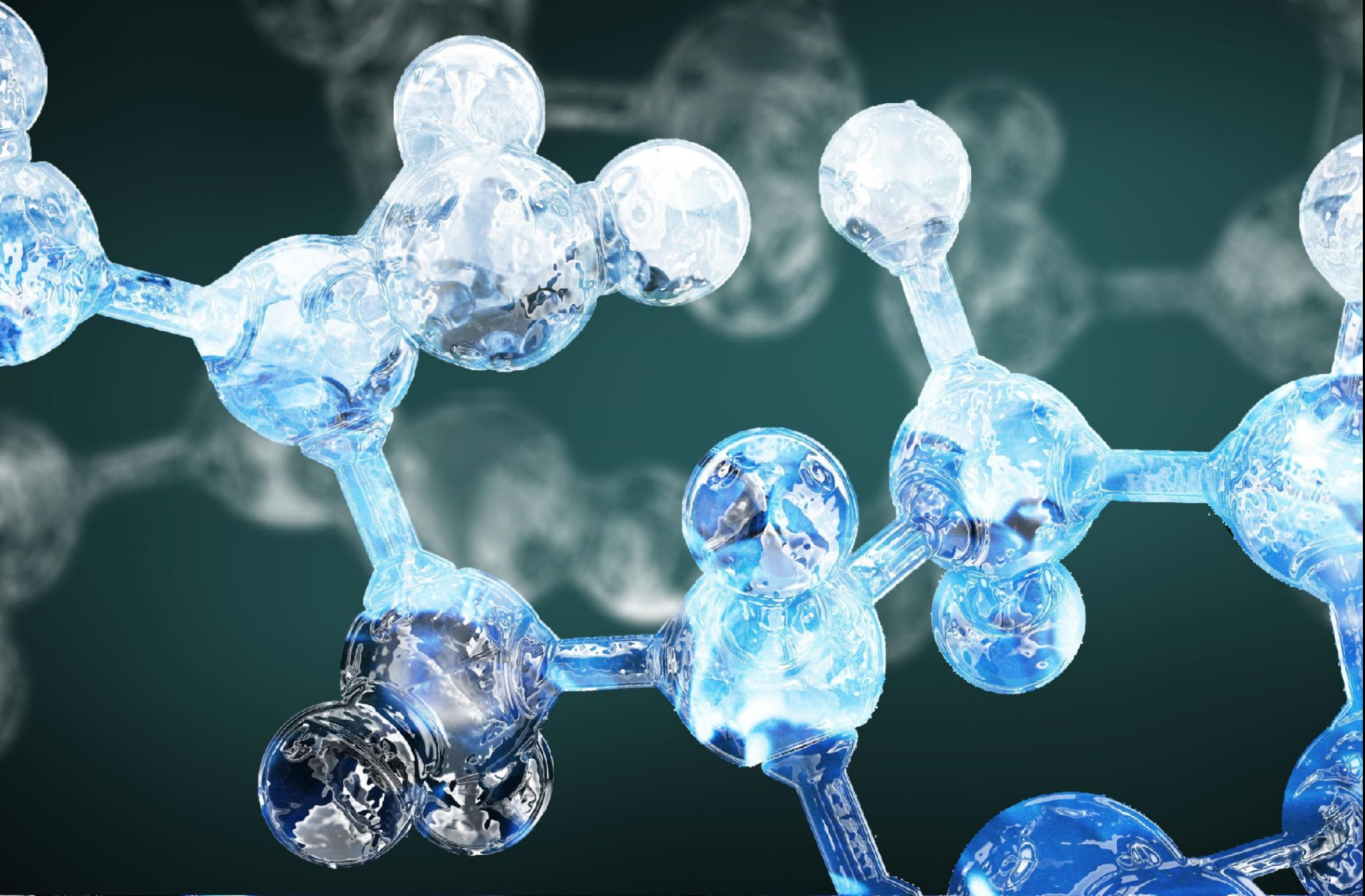


Molecular Compounds



Learning Outcomes:

CC1.33 Recall examples of common covalent, simple molecular compounds.

CC1.33 Describe the general properties of covalent, simple molecular compounds.

CC1.33 Explain why covalent, simple molecular compounds have low melting and boiling points.

CC1.33 Explain why covalent, simple molecular compounds are poor conductors of electricity.

CC1.38 Describe the structure of a polymer.

IONIC or COVALENT?

1. I am a compound that is held together by electrostatic force due to my oppositely charged components.
2. I am a compound made through the gaining and losing of electrons.
3. I am a compound where my components share electrons.
4. I am a compound where my components now have a full outer shell of electrons.

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IONIC or COVALENT

Picture puzzles-

deduce the following key words from the pictures

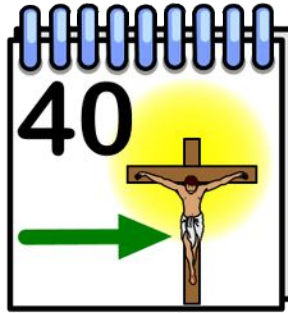
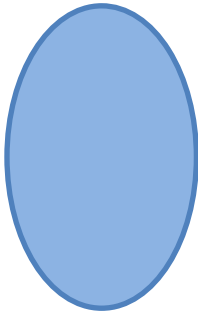
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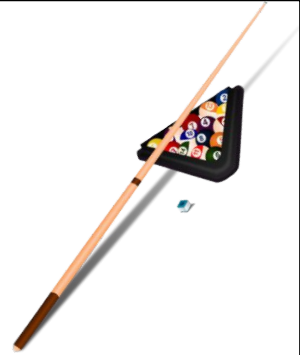
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3



4



5



ANSWERS

1. Compound
2. Bonds
3. Covalent
4. Molecule
5. Element



All compounds contain atoms of more than one element, chemically joined together by bonds.

Some compounds exist as molecules- distinct groups of atoms joined by covalent bonds. They have covalent, simple molecular structures.

An example is _____, where one molecule always has one atom of oxygen bonded to two atoms of hydrogen.

KEYWORDS:

Bonds



Hydrogen

Compounds

Covalent

CHALLENGE: What do you think might be the properties of covalently bonded compounds?

Simple Molecular Compounds



Intermolecular Forces

Covalent bonds are very **strong**. The atoms within each molecule are held very tightly together. But **forces of attraction** between **molecules** is small.

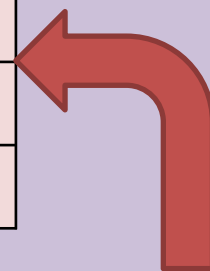
- there are weak intermolecular forces between molecules. Overcoming these forces does not take much energy.

Small simple molecules such as water often have low melting and boiling points because it doesn't take much energy to overcome the weak **intermolecular forces**.

Investigating Melting Points

Look at the following data:

Material	Melting Point (°C)	Formula
octadecane	29	$C_{18}H_{38}$
docosane	42	$C_{22}H_{46}$
heptacosane	59	$C_{27}H_{56}$
dotriacontane	69	$C_{32}H_{66}$



When we add in their molecular formulae, what do you notice:

TASK: Describe the pattern that you see. Then try to explain it.



Model answer

As the number of carbons and hydrogens in the material increases so does the melting point. This is because the longer the chain of carbon and hydrogen atoms, the more energy is required to break the bonds and cause the material to change state to a liquid.

In **Green pen**, mark your answer:

- How could you improve your answer?
- Did you describe AND explain like the question asked?
- Was your explanation correct?

RECAP: Compounds that are ionically bonded can conduct electricity when molten/dissolved because they have positive and negative ions that can carry a charge and move around.

Knowing what we know about ionic bonding, make a prediction about whether simple covalent compounds can conduct electricity.

End

2 minutes

Team 'simple covalent compounds
cannot conduct electricity'

WHY?



COVALENT BONDING
EDITION!!

WHY?

Team 'simple covalent compounds
can conduct electricity'

Team 'simple covalent compounds cannot conduct electricity'

An electric current is a flow of charged particles. Simple molecules have no overall charge so cannot carry an electric current.

In a covalent bond, electrons are shared between two atoms. The strong forces between negatively charged electrons and the positively charged nuclei hold the electrons in place. The electrons cannot flow so cannot carry a current.

Use the above information to explain why water is a poor conductor of electricity.

Polymers



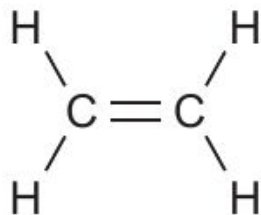
Polymer means: "many parts"

Polymers are long chains of molecules and they can be cross-linked, joining several chains together.

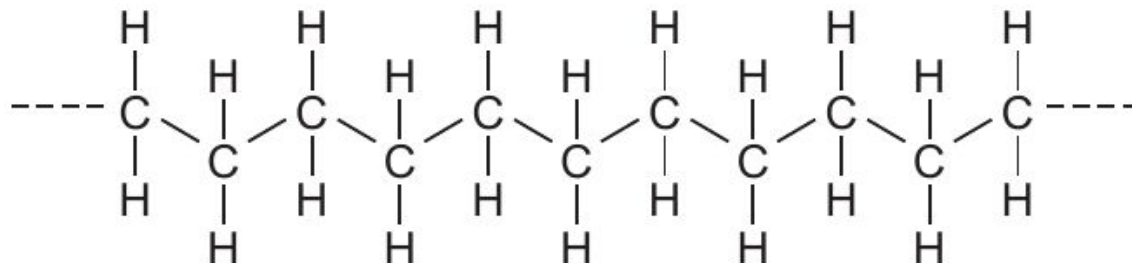
They are made from smaller molecules called **monomers** (mono = one).

Jelly, rubber bands, plastic bags and chewing gum are examples of polymers.

(i) ethene



(ii) poly(ethene)



Try to explain what **polymers** and **monomers** are using ethene and polythene as an example.

Use the keywords to help you.

CHALLENGE: Add what you now know about polymers to your previous answer on melting and boiling points.

KEYWORDS:



Polymer

Monomer

Molecules

Chain

Linked

Smaller

Ethene

Polythene